

Nermine Cheriaa

AI Engineering Student — Generative AI & Agentic Systems

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Engineering student specializing in Generative AI and Agentic Systems, with hands-on experience designing RAG-based conversational agents, VLM-powered document analysis pipelines, and multimodal AI systems end-to-end – from model integration to production deployment (FastAPI, Docker, Hugging Face). Currently seeking an End of study project internship to build production-grade GenAI and agentic solutions.

Professional Experience

Generative AI Intern — Talan, Tunis Jul 2026 – Aug 2026

SANTAID – AI-powered Health Insurance & Fraud Detection Platform

- Developed an AI-powered assistant based on Retrieval-Augmented Generation (RAG) for health insurance reimbursements and administrative guidance.
- Designed a hybrid fraud detection pipeline (supervised ML, Isolation Forest, business rules) enhanced with a Knowledge Graph to detect suspicious networks between patients and healthcare professionals.
- Integrated an AI investigation copilot using Mistral AI to explain fraud alerts and generate investigation summaries.

AI Intern — Hydatis, Tunis Jun 2025 – Aug 2025

- Developed AI-driven conversational solutions within a smart security platform.
- Improved NLP performance and optimized conversation flows.
- Integrated chatbot with APIs and databases in a microservices architecture.

Projects

Agentic Research Copilot – Corrective RAG with Multi-Agent Verification Aug 2026

LangGraph, Next.js, Qdrant, RAGAS, Docker, GitHub Actions

[GitHub Repo](#)

- Designed and deployed a self-correcting RAG agent (LangGraph) with real-time streaming UI, dynamically triggering query rewriting and web search fallback via LLM-based relevance grading.
- Built a multi-agent verification loop (generator vs. critic) and measured faithfulness/context precision improvements using RAGAS evaluation.
- Shipped to production with CI/CD (GitHub Actions) and a public live demo (Qdrant, Docker, Vercel/Railway).

Handwritten Student Feedback Analysis – VLM-Based Document Analysis Jul 2026

Gemini Vision, VLM, FastAPI

[GitHub Repo](#)

- Built an end-to-end pipeline using Gemini Vision to transcribe handwritten forms, extracting and classifying individual expectations by category and sentiment (Positive/Negative).
- Delivered both an interactive single-document interface and a batch processing pipeline outputting structured JSON.

Sentiment Analysis API – NLP End-to-End Mar 2026

HuggingFace, FastAPI, Docker, Streamlit

[GitHub Repo](#)

- Fine-tuned DistilBERT on IMDb dataset (3k samples), achieving 89.4% accuracy and 0.896 F1-score.
- Exposed model via FastAPI REST endpoints supporting single-text and batch CSV inference.

Education

Engineering Degree in Computer Networks and Telecommunications (*Master's Equivalent*) 2022 – Present

National Institute of Applied Sciences and Technology (INSAT)

Baccalaureate in Computer Science 2022

Grade: 18.04/20 – Highest Honors

Skills

LLM & Agentic AI: RAG, LangGraph, RAGAS, Mistral AI, Gemini, Hugging Face Transformers

AI/ML: Scikit-learn, XGBoost, PyTorch, NLP

Programming: Python, C/C++, Java

Backend: Spring Boot, FastAPI

Databases: SQL, MongoDB, Qdrant

Tools: Git, Docker, Hugging Face

Certifications

- Efficient Large Language Model Customization – NVIDIA
- CCNAV7: Introduction to Networks – CISCO
- CCNAV7: Switching, Routing, And Wireless essentials – CISCO
- Cyber Security Essentials – CISCO

Languages

English: Professional (B2)

French: Fluent (C1)

Arabic: Native